

INTRODUCTION

Substances present around us are classified under two categories—pure substances and impure substances.

- Substances which are made up of the same kind of atoms or molecules are called **pure substances**. Pure substances are further classified into elements and compounds.
- Substances which are made up of two or more pure substances physically mixed together in any ratio are called **impure substances**. Mixtures are impure substances.

Let us learn more about pure and impure substances.

IDENTIFICATION OF ELEMENTS

Elements are the simplest pure substances that cannot be broken down into two or more simpler substances by any physical or chemical method. There are 118 elements that have been identified, out of which the first 94 occur naturally on the Earth and the remaining 24 elements have been prepared artificially. Some of the common elements are hydrogen, oxygen, nitrogen, carbon, iron, copper, sodium, zinc, aluminium, sulphur, mercury, silver and gold.

All the atoms of a particular element are identical but the atoms of one element differ from that of another element. For example, all the atoms of oxygen are identical and all the atoms of nitrogen are identical. However, the atoms of oxygen are different from the atoms of nitrogen.

SCIENCE ALIVE



Information Literacy
Understanding Facts

- Calcium is an important element, which helps build our bones.
- Iron is present in the haemoglobin of red blood cells which carry oxygen to various parts of the body.
- Plants contain magnesium in chlorophyll, which is essential for photosynthesis.

REASON CORNER



Critical Thinking
Reasoning, Applying

Which elements do you think are most important for our survival? Give three examples with reasons.

REPRESENTATION OF ELEMENTS

The elements are represented using short notations called symbols. A symbol is a short representation of the name of an element. The International Union of Pure and Applied Chemistry (IUPAC) approves the names of the elements.

Some elements are represented by the first letter of their English name (written in capitals). For example, oxygen is represented by the letter O, nitrogen is represented by the letter N.

When the first letter of several elements is same, then the elements are represented by two letters. The first letter of the symbol is written in capital and the second letter is written in small. For example, both calcium and cobalt have C as their first letter. So, calcium is represented by Ca and cobalt is represented by Co. Similarly, chlorine is represented by Cl and chromium is represented by Cr.

Symbols of some elements are derived from their Latin names. For example, Na is the symbol of sodium whose Latin name is *natrium* and K is the symbol of potassium whose Latin name is *kalium*. A symbol represents one atom of an element. For example, the symbol C represents one atom of carbon and the symbol H represents one atom of hydrogen.

Table 3.1 lists the names, Latin names and symbols of some elements.

Table 3.1 Symbols of some elements

Element	Latin name	Symbol	Element	Latin name	Symbol
Aluminium	—	Al	Helium	—	He
Antimony	Stibium	Sb	Hydrogen	—	H
Argon	—	Ar	Iodine	—	I
Arsenic	—	As	Iron	Ferrum	Fe
Barium	—	Ba	Krypton	—	Kr
Boron	—	B	Lead	Plumbum	Pb
Bromine	—	Br	Lithium	—	Li
Calcium	—	Ca	Manganese	—	Mn
Carbon	—	C	Magnesium	—	Mg
Chlorine	—	Cl	Mercury	Hydrargyrum	Hg
Chromium	—	Cr	Nickel	—	Ni
Cobalt	—	Co	Nitrogen	—	N
Copper	Cuprum	Cu	Neon	—	Ne
Fluorine	—	F	Oxygen	—	O
Germanium	—	Ge	Phosphorus	—	P
Gold	Aurum	Au	Platinum	—	Pt

Potassium	<i>Kalium</i>	K	Sulphur	—	S
Radium	—	Ra	Tin	<i>Stannum</i>	Sn
Radon	—	Rn	Tungsten	<i>Wolframium</i>	W
Silicon	—	Si	Uranium	—	U
Silver	<i>Argentum</i>	Ag	Xenon	—	Xe
Sodium	<i>Natrium</i>	Na	Zinc	—	Zn

IDENTIFICATION OF COMPOUNDS

Compounds are the pure substances made up of two or more elements chemically combined together in a fixed ratio. When two or more elements are combined, they lose their individual properties and a new compound is formed which has a different set of properties from the elements it is composed of.

SCIENCE TALK

Form groups of 4-5 students and discuss the following questions. Present your views in class.

What changes must happen for elements to become a compound? Would there be life on Earth if elements did not form compounds? Why?



• Critical Thinking • Communication
• Reasoning, Analysing • Discussing, Speaking

SCIENCE ALIVE

- Silver chloride, a compound, is used in photographic films.
- Compounds of phosphorus, nitrogen and potassium are used as fertilisers (Fig. 3.1).
- Sodium fluoride is the most commonly used compound in toothpastes as it strengthens the teeth.
- Sugar and aspirin (a medicine) are both compounds of carbon, hydrogen and oxygen, combined in different proportions.



Fig. 3.1 Fertiliser containing nitrogen, phosphorus and potassium



• Information Literacy
• Understanding Facts

Table 3.2 lists some compounds and their formulae.

Table 3.2 Some compounds and their formulae — In atomic structure.

Compound	Constituents	Formula
Carbon dioxide	Carbon and oxygen	CO ₂
Water	Hydrogen and oxygen	H ₂ O
Sodium chloride (common salt)	Sodium and chlorine	NaCl
Sand	Silicon and oxygen	SiO ₂
Methane	Carbon and hydrogen	CH ₄
Ammonia	Nitrogen and hydrogen	NH ₃
Calcium oxide	Calcium and oxygen	CaO
Sodium sulphate	Sodium, sulphur and oxygen	Na ₂ SO ₄
Sodium carbonate	Sodium, carbon and oxygen	Na ₂ CO ₃
Plaster of Paris (hydrated calcium sulphate)	Calcium, sulphur, hydrogen and oxygen	CaSO ₄ · 2H ₂ O
Sodium hydrogen carbonate	Sodium, hydrogen, carbon and oxygen	NaHCO ₃

REASON CORNER

Substance 'A' on heating breaks down into two substances, 'B' and 'C'. Is A an element or a compound? Give reasons.



• Critical Thinking
• Reasoning, Analysing

MIXTURES

Mixtures are the impure substances which contain two or more kinds of particles (elements or compounds or both) mixed together in any ratio. A mixture is formed when two or more substances are mixed such that their particles are in intimate contact with one another without undergoing a chemical change.



Fig. 3.2 Some examples of mixtures

For example, lemonade is a mixture of lemon juice, sugar and water; air is a mixture of nitrogen, oxygen, carbon dioxide, water vapour and other gases. Some other examples of mixtures (Fig. 3.2) are tea, honey, milk, ice cream and toothpaste.

SCI-VOCAB

mixture: an impure substance which contains two or more kinds of particles (elements or compounds or both) physically mixed together in any ratio

Differences between compounds and mixtures

Both compounds and mixtures are made up of two or more substances but they are different from each other.

Let us study the differences between compounds and mixtures listed in Table 3.3.

Table 3.3 Differences between compounds and mixtures

Compounds	Mixtures
They are pure substances.	They are impure substances.
A compound involves a chemical reaction between two or more substances.	A mixture is formed by the physical combination of two or more substances.
The substances are combined in a certain fixed ratio.	The ratio of the substances of a mixture is not fixed. They can be mixed in any ratio.
The properties of a compound are different from its components.	In a mixture, the properties of the components are retained.
Compounds cannot be broken down to their individual components through physical means.	Mixtures can be separated into their individual components through physical means.
Examples: Water and rust	Examples: Honey and milk

MYTH VS FACT

Myth: Plant-protection products containing copper compounds are harmful for both the plants and the equipment used for gardening.

Fact: The copper compounds do not harm the gardening equipment as long as good agricultural practices are followed. Also, the copper compounds are observed to be harmful only for selected plant types, depending on how sensitive the plant is and in which season these compounds are sprayed.

REASON CORNER

Air is a mixture because its components, that is, oxygen, nitrogen, etc., do not react with each other. True or false?

SCIENCE ALIVE

Blood is a mixture (Fig. 3.3) of various components such as platelets, red blood corpuscles, white blood corpuscles and plasma. Blood can be separated into its components in a device called **centrifuge**.

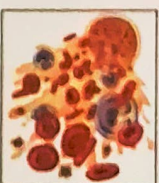


Fig. 3.3 Blood—a mixture

ACTIVITY



UNDER ADULT SUPERVISION



• Experiential Learning
• Concrete Experience

Aim: To study the difference between compounds and mixtures

Materials required: Sulphur powder, iron filings, a bar magnet, two china dishes and a burner

Procedure: Take a small quantity of sulphur powder and iron filings in a china dish and mix them well. Now bring a bar magnet above the china dish.

Observation: You will observe that the iron filings are attracted towards the magnet (Fig. 3.4).

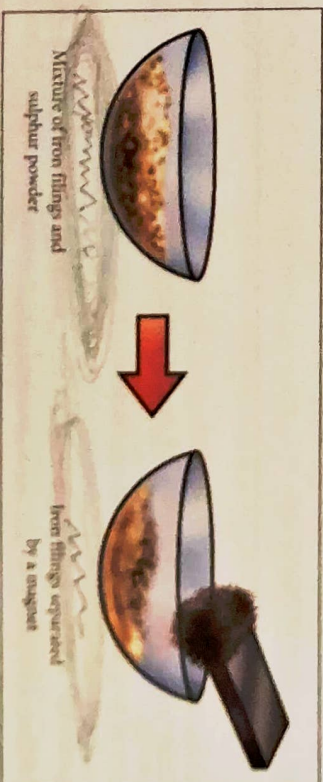


Fig. 3.4 Iron filings are attracted towards the magnet and get separated.

Procedure: Now, take 8 g of sulphur powder and 14 g of iron filings in another china dish and mix them well. Heat the mixture on the burner. Now, bring the bar magnet above the china dish.

Observation: You will observe that the substance is not attracted by the magnet (Fig. 3.5).

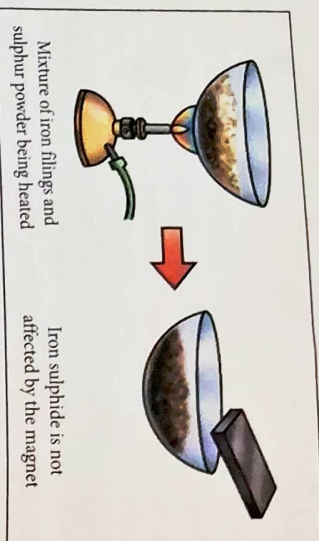


Fig. 3.5 Compound iron sulphide is not affected by the magnet.

Conclusion:

1. When iron and sulphur are mixed in any quantity, a mixture is obtained where iron retains its original property of getting attracted towards a magnet and thus can easily be separated from the mixture. Sulphur powder left behind retains its yellow colour and characteristic smell.
2. When iron and sulphur are mixed in a fixed quantity and heated, a compound called iron sulphide is formed which has different properties from those of iron and sulphur. It is no more a powder. It is a dark, dense mass. It neither gets attracted towards the magnet nor possesses the characteristic smell of sulphur. It is not even possible to separate the components of the compound iron sulphide by any physical method of separation.

Types of mixtures (on the basis of composition)

The two types of mixtures on the basis of composition are as follows:

- ❖ **Homogeneous mixtures:** A mixture with uniform composition in which the components cannot be seen separately is called a homogeneous mixture. It has a uniform composition and every part of the mixture has the same properties. Tap water, milk, air, ice cream, sugar solution, acetic acid in water, steel and bronze are some of the examples of homogeneous mixtures.

- ❖ **Heterogeneous mixtures:** A mixture which does not have a uniform composition and in which the components can be seen separately is called a heterogeneous mixture. It does not have a uniform composition and properties. A mixture of salt and sand, a mixture of oil and water, and a mixture of chalk powder and water are some of the examples of heterogeneous mixtures.

Types of mixtures (on the basis of state of the components)

Table 3.4 lists the examples of different types of mixtures on the basis of the state of their components.

Table 3.4 Examples of different types of mixtures

State of components	Type of mixture	Examples
Solid-Solid	Homogeneous	Brass, steel and bronze: mixtures of metals
	Heterogeneous	Sugar and salt; stones and rice
Solid-Liquid	Homogeneous	Sugar and water; salt and water
	Heterogeneous	Chalk and water; sand and water
Liquid-Liquid	Homogeneous	Water and lemon juice; water and alcohol
	Heterogeneous	Oil and water
Liquid-Gas	Homogeneous	Aerated water (aerated water or soda is a mixture of carbon dioxide dissolved in water)
	Heterogeneous	Mist (mixture of water droplets and air)
Solid-Gas	Heterogeneous	Smoke (carbon particles in air)
Gas-Gas	Homogeneous	Air (mixture of gases)

The liquids which completely dissolve in each other and make homogeneous mixtures are called **miscible liquids**. For example, milk and water are miscible liquids.

The liquids which do not dissolve in each other completely and make heterogeneous mixtures are called **immiscible liquids**. For example, oil and water are immiscible liquids.

LIFE SKILLS



- Experiential Learning • Critical Thinking
- Global Citizenship Skills • Life Skills • Communication
- Active Experimentation • Reasoning, Inferring
- Environment Literacy, Health Literacy • Initiative • Discussing

Collect some samples of solid-solid, solid-liquid and liquid-liquid types of mixtures from your daily life and classify them as homogeneous or heterogeneous. Are mixtures important in our life? Discuss.

What happens to the pure substances and mixtures that we create? How can we safely dispose household toxic wastes such as batteries, paint and cleaners?

The discussion to answer these questions is aligned with the United Nations' Sustainable Development Goal 3—Good Health and Well-Being.



ACTIVITY



• Experiential Learning • Critical Thinking
• Concrete Experience • Inferring

Aim: To differentiate between different types of mixtures

Materials required: Sugar, handful of rice, common salt, water, cooking oil, stirrer, a beaker and a china dish

Procedure: Perform the following activities and classify the mixtures as homogeneous or heterogeneous.

1. Take some rice and sugar in a china dish and mix them by hand.
2. Take water in a beaker and add common salt in it. Stir it well to make salt solution.
3. Take some water and cooking oil in a china dish. Mix them well.

Observation and conclusion: Complete the table with your observation and then conclude whether the mixtures are homogeneous or heterogeneous.

Mixture	Type of mixture
Mixture 1—rice and sugar	Homogeneous
Mixture 2—salt and water	Homogeneous
Mixture 3—oil and water	Heterogeneous

ECO-SOLDIER



• Information Literacy • Art-integrated Learning
• Global Citizenship Skills
• Research Skills • Collage-making • Environment Literacy

You know that oil and water do not mix. Due to human activity, at times, petroleum spills on marine water from the tankers carrying it. Find out methods by which oil which has spilt over marine water can be removed. Present your findings through a collage.

MY QUESTION TIME



• Critical Thinking • Communication
• Framing Questions • Discussing

Frame a question on types of mixtures and discuss its answer.

CHECK YOUR PROGRESS



• Critical Thinking
• Remembering, Applying

Tick (✓) the correct option.

1. Common salt is represented by the formula
a. NaCl ✓ b. NaCl_2 c. Na_2Cl
2. The symbol of phosphorus is derived from its
a. Latin name ✓ b. English name. c. first and last syllables. ✓
3. A mixture in which the components cannot be seen separately is called a
a. heterogeneous mixture. b. clear mixture. c. homogeneous mixture. ✓

PREPPING FOR PISA



• Critical Thinking • Global Citizenship Skills
• Comprehension, Reasoning • Health Literacy

Read the passage and answer the questions that follow.

Mercury is a rare element in the Earth's crust (Fig. 3-6). It has many uses. It is used in thermometers, dental amalgams, liquid mirrors and fluorescent lamps. However, exposure to even a small amount of mercury can be hazardous. Moreover, mercury does not degrade over time. When discharged into water or land, it evaporates into the atmosphere (mercury vapours) where it can travel for up to thousands of kilometres before it comes down through rainfall and deposits into the oceans or land.

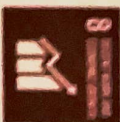
This mercury can reach us when we eat contaminated seafood. Exposures may also result from using or breaking products containing mercury, and making and handling objects that contain mercury. Mercury can adversely affect the central and peripheral nervous systems. Thus, extreme caution must be taken while handling mercury, and proper equipment must be provided to the industrial workers whose work requires exposure to mercury.



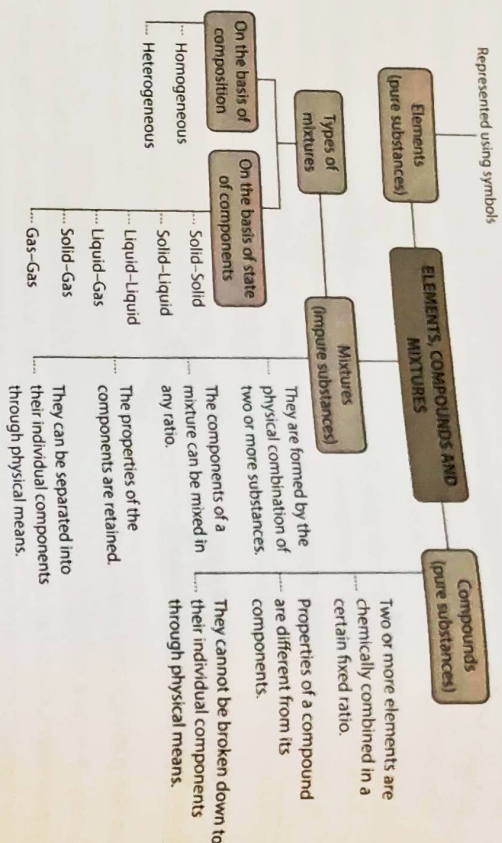
Fig. 3-6: Mercury metal on an ampoule

1. Which of the following would be most affected by breathing mercury vapours?
a. heart b. brain c. ears d. eyes
2. State whether True or False (T/F).
a. Mercury is poisonous only when food containing mercury is eaten.
b. Mercury remains in the environment for a long period of time.
3. Explain why it is advised not to keep mercury thermometers under the tongue of children to measure their body temperature.

The answers to these questions is aligned with the United Nations' Sustainable Development Goal 8—Decent Work and Economic Growth



CONCEPT RECALL



EVALUATION

A. Choose the correct option.

- C is the symbol of b. chlorine. c. carbon. d. cobalt.
- NaHCO_3 is the formula of a. sodium carbonate. b. sodium hydrogen carbonate. c. sodium oxide. d. sodium hydroxide.
- A mixture of coconut powder and water is a a. liquid-liquid mixture. b. solid-solid mixture. c. solid-liquid mixture. d. liquid-gas mixture.
- Plaster of Paris is a compound of a. sulphur, calcium, oxygen and hydrogen. b. calcium, phosphorus and hydrogen. c. calcium, sulphur, carbon and oxygen. d. hydrogen, sulphur and oxygen.
- The chemical equation given below represents the burning of methane.
 $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
 Identify the element in the equation.
a. CH_4 b. O_2 c. CO_2 d. H_2O

HOTS

B. Fill in the blanks.

- A is an impure substance.
- The Latin name of Iron is
- A is the short representation of an element.
- Compounds of, and are used as fertilisers.
- Carbon and hydrogen are the constituents of

C. State whether True or False (T/F).

- Liquids which do not dissolve in each other are miscible liquids. ☐
- A mixture of lemon juice and water is a homogeneous mixture. ☐
- Compounds can be easily broken down into their components by physical means. ☐
- Silver chloride is used in photographic films. ☐

D. Give one word for the following:

- chemical name of common salt
- an element present in chlorophyll of plants
- a device that can separate the components of blood

E. Short-answer-type questions

- Write the symbols of the following elements:
 boron, ~~germanium~~, ~~tin~~, platinum, sodium, iron, ~~manganese~~
~~copper~~, ~~zinc~~
- Name the following compounds: SiO_2 , CH_4 , $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, NH_3
~~magnesium~~
- Give two examples each of miscible liquids and immiscible liquids.

F. Long-answer-type questions

- How are the symbols of elements derived?
- Define mixture. Explain different types of mixtures.
- Differentiate between mixtures and compounds.

Higher Order Thinking Skills (HOTS)

Two substances, X and Y, react to form a third substance X_2Y (product). Which of the following statements concerning the product are true?

Underline the correct answer.

- The product so formed is an element.
- The product so formed is a compound.
- The product will always have a fixed composition.
- The product shows the properties of substances X and Y.

21st CENTURY SKILLS
• Thinking
• Analysing, Applying
Reasoning

Learning by Doing

Some of the heterogeneous mixtures we consume often are aerated drinks. Aerated drinks have no nutritional value, and have sugar and chemicals dissolved in them.

Form groups of 4-5 students. Make a poster to raise awareness about the harmful effects of aerated drinks on human health.



- Collaboration • Art-integrated Learning
- Global Citizenship Skills
- Group Work • Poster-Making • Health Literacy

Science and Scientists

Robert Boyle (1627–1691) was an Irish philosopher, chemist, physicist and inventor. He coined the term 'element' in the year 1661. He endorsed the view that elements cannot be decomposed. He further supposed that the elements were ultimately composed of particles of various sorts and sizes, into which, however, they were not to be resolved in any known way. He also made the distinction between mixtures and compounds. He did considerable work in the technique of detecting the constituents of compounds. Boyle also tried to answer complex religious questions using methods derived from his scientific practices.



- Information Literacy
- Understanding Fac

